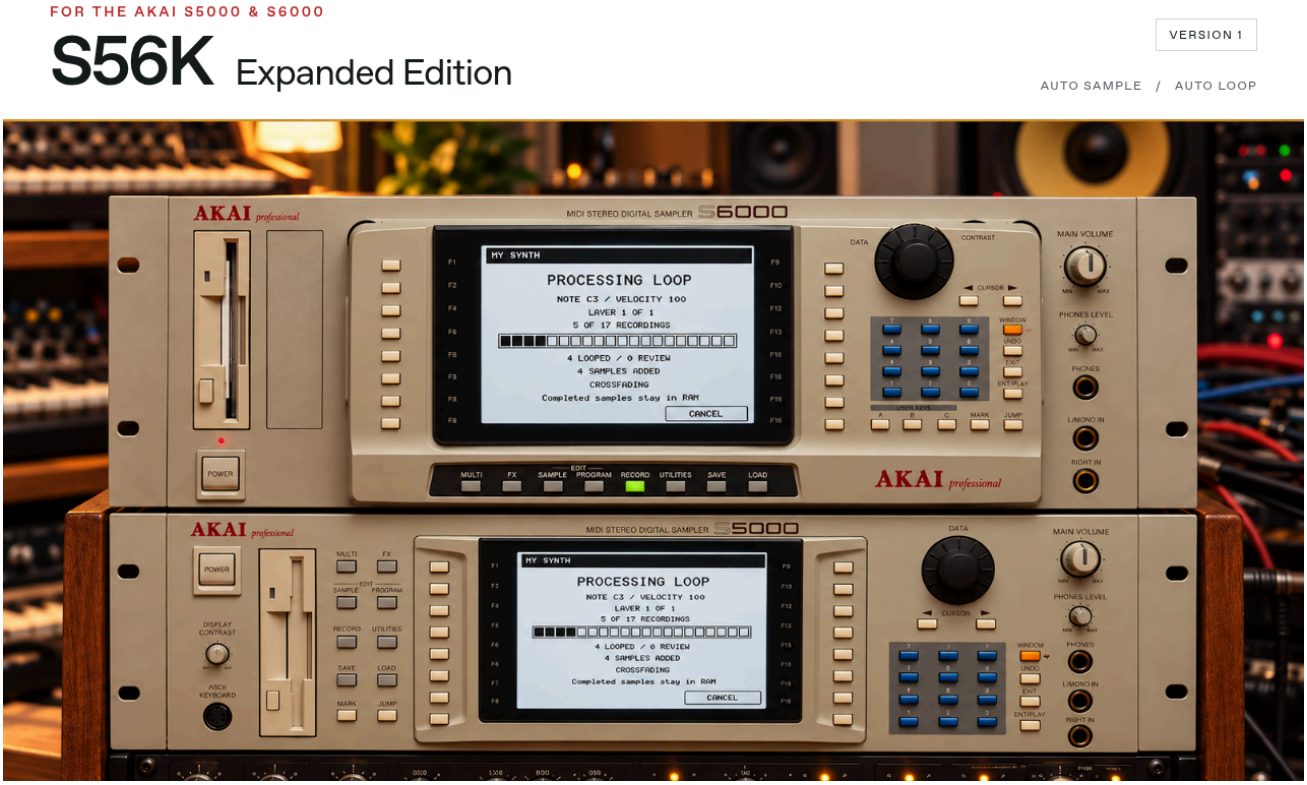
The user guide.

Auto Sample & Auto Loop
For the AKAI S5000 and S6000



Create. Capture. Play.
Turn an external instrument into a playable Akai program. Choose your notes and velocities, prepare the sustain, and save when you are ready.

Read the installation guidance, compatibility evidence and flashing disclaimer in the appendices before loading firmware.

Version 1 · 22 October 2026 · Independent project for AKAI hardware

Inside the guide

Create and name an empty program. Capture your instrument into RAM. Listen, refine and save the program with its samples through the normal SAVE page.

01 Connect and prepare

02 Set the capture range

03 Check levels

04 Capture velocity layers

05 Recording and Auto End

06 Loop the sustained part

07 Run, listen and save

08 Troubleshooting

09 Expanded Edition startup

Installation and reference

A · Installation B · Compatibility C · Flashing disclaimer

About this edition

S56K Expanded Edition adds automatic sampling and optional loop preparation to the AKAI S5000 and S6000. The visible EXPANDED EDITION v1 label identifies these additions; the underlying OS identity remains 2.14.

The screen illustrations use native firmware renders with example settings and progress states. The studio artwork is a presentation image, not evidence of hardware testing.

Firmware availability and exact-file instructions are published separately in the release area. This manual does not contain an OS image.

YOUR PROGRAM. YOUR WORKFLOW.

01

Create & name.

Start with an empty program.
Choose the sound to capture.

02

Capture into RAM.

Record, map and optionally loop.
Listen across the finished program.

03

Save when ready.

Use the normal SAVE page.
Your samples stay under your control.

1. Connect and prepare

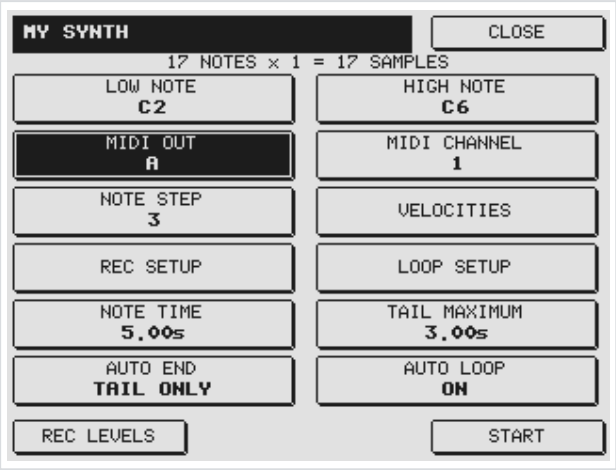
Connect the sampler's **MIDI OUT A or B** to the instrument's MIDI input. Connect the instrument's audio output to the sampler's recording input: left input for a mono capture, or both left and right for stereo. Select the appropriate input source using the normal recording controls.

Choose the sound on the instrument, set its receiving MIDI channel and leave that sound selected throughout the run.

Create an **empty program**, give it a useful name and open **Program Tools → Auto Sample**. The initial empty keygroup must contain no sample assignments. A run adds its own keygroups and velocity zones. To make a second run after completing or cancelling the first, start with another empty program.

Leave generous free sample RAM. Stereo recordings and multiple velocity layers use more memory, and loop processing needs temporary copies while preserving the original recording. During a run, avoid editing samples simultaneously through ak.Sys or SysEx.

2. Set the capture range



Press the soft key beside a field to select it, then use the **DATA wheel** or **+ / –** to change the value. The submenu buttons open their own pages; **BACK** returns to the main setup.

Main-page control	What it does
LOW NOTE / HIGH NOTE	First and last pitches to capture. Set both to the same note for a one-note test.
MIDI OUT	Selects physical output **A** or **B**, matching the rear-panel labels.
MIDI CHANNEL	Must match the instrument's receiving channel.
NOTE STEP	Distance between captured notes, in semitones. Smaller steps produce more recordings and less pitch shifting during playback.
NOTE TIME	Requested time to hold each outgoing note before sending Note Off.
TAIL MAXIMUM	Maximum recording time after Note Off, allowing the instrument's release and effects to fade.
AUTO END	Chooses whether quiet audio can shorten the held note, the tail, or neither.
AUTO LOOP	Enables optional loop processing after each recording.

The summary shows pitches × velocity layers = recordings. The high endpoint is included even when the final interval is shorter than Note Step. The limit is **99 captured pitches**, with up to four layers per pitch, subject to available memory.

Note names follow the Akai convention: **MIDI note 60 is C3**. Another instrument may call the same pitch C4; match the MIDI note number.

For example, C2 to C4 with a step of 12 captures C2, C3 and C4. With four velocity layers, that produces twelve recordings. The sampler assigns playback ranges between neighbouring captured pitches across the selected low-to-high range.

3. Check levels and choose mono or stereo

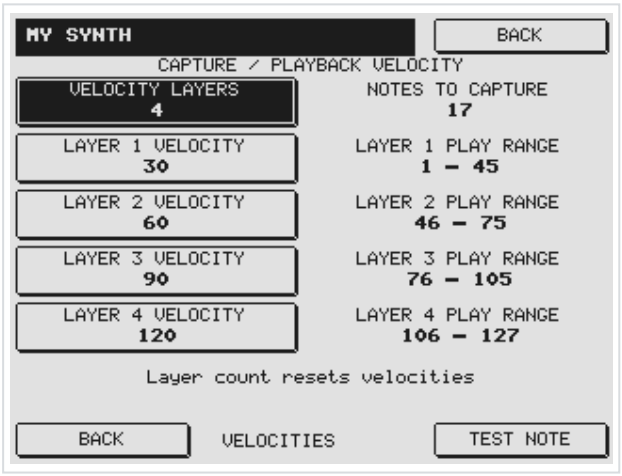
Open **REC LEVELS** before starting a long run. This uses the sampler's existing input, gain and meter controls. Set **Record Mode** to MONO or STEREO here; the choice is retained when returning to Auto Sample.

Use **TEST NOTE** and inspect the meters at the loudest velocity you plan to capture. Adjust the instrument's output and the sampler's recording gain for a healthy signal without clipping. For stereo, check both channels.

Velocity may change the instrument's tone as well as its loudness. Recording gain controls the captured level. Set both appropriately for the sound.

START LEVEL, available in REC SETUP and through the native recording controls, determines when incoming audio triggers recording. Set it above the background noise but low enough to catch the intended attack. A slow or very quiet sound may need a lower threshold. Test one note and inspect its beginning before recording the entire range.

4. Capture velocity layers



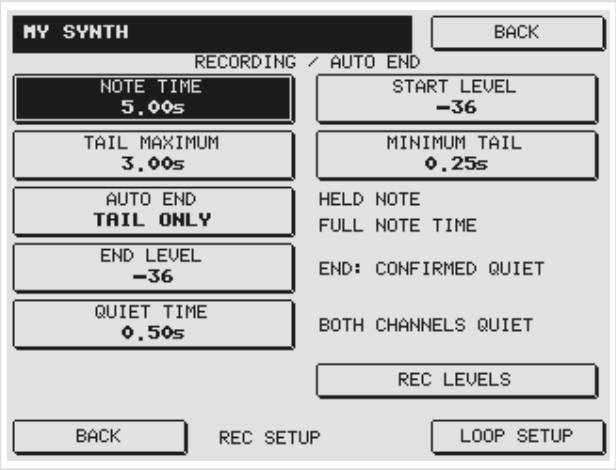
Open **VELOCITIES** and choose one to four layers. Each layer specifies the MIDI velocity sent to the instrument. The displayed **PLAY RANGE** shows which incoming velocities will select that layer when you play the completed Akai program. Only the selected number of layers is displayed: with one layer, the other three rows are hidden.

Changing the layer count resets the capture velocities to a starting arrangement. Adjust them afterwards to suit the instrument. Velocities must increase from layer to layer; the controls maintain that order.

With four layers, the starting values are 30, 60, 90 and 120. Their playback ranges are divided at the midpoints, covering velocities 1–127 without gaps. Select a layer and use **TEST NOTE** to audition that velocity at the current low note.

Each pitch gets one keygroup, with its recordings placed in the appropriate velocity zones. Sample names include a shortened program-name prefix, the note and the capture velocity. Longer names are shortened to fit the sampler's naming limit.

5. Decide when recording should end



NOTE TIME covers the held note. **TAIL MAXIMUM** covers the sound after the external instrument receives Note Off. A sustained pad may keep sounding while held, then take several seconds to release. A guitar or piano sound may decay to silence while the note is still held.

AUTO END setting	Held note	Release tail
OFF	Uses the full Note Time.	Uses the full Tail Maximum.
TAIL ONLY	Uses the full Note Time.	Can stop early after confirmed quiet.
NOTE + TAIL	Can send Note Off early after the sound has been heard and then becomes quiet.	Can also stop early after confirmed quiet.

For sustained sounds that you intend to loop, **TAIL ONLY** is a useful starting point. It preserves the requested held duration while avoiding an unnecessarily long silent release. For naturally decaying sounds, **NOTE + TAIL** can save time across a large run.

When Auto End is enabled, REC SETUP exposes:

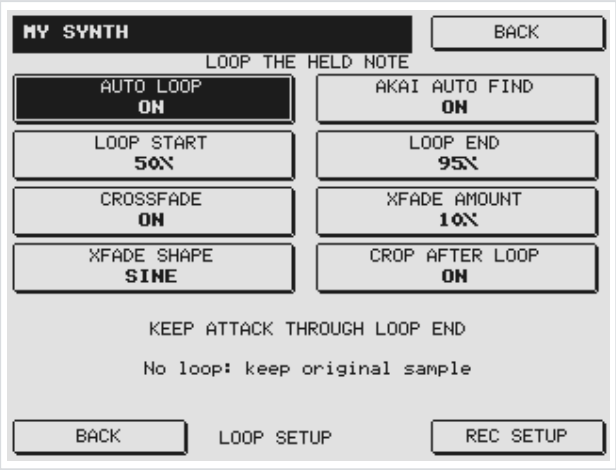
- **END LEVEL:** the level at or below which audio counts as quiet.
- **QUIET TIME:** how long that quiet condition must persist before it qualifies as an ending.
- **MINIMUM TAIL:** the earliest point after Note Off at which quiet can end the tail.
- **MINIMUM NOTE:** available in NOTE + TAIL mode; prevents quiet detection from shortening the held note before this time.

In stereo, **both channels must be quiet**. This helps retain a release or effect that continues on only one side. Longer Quiet Time is useful for sounds with rhythmic gaps, tremolo or delays.

The end detector uses the sampler's coarse meter scale. **METER FLOOR** is its lowest detection bucket, not a precision dBFS measurement. Choose a setting by listening to a test recording and observing the actual noise floor. A lower end threshold may preserve more decay, but noise can prevent it from being reached.

Unavailable controls disappear. Their values remain stored and reappear when the relevant mode is enabled. Auto End changes recording behaviour; it does not switch Auto Loop on or off.

6. Loop the sustained part



Enable **AUTO LOOP** and open **LOOP SETUP**. Its percentages refer to the **captured held-note portion**, starting at the sample's playback start and ending before the external release tail. They do not include the tail in their measurement.

Loop control	Meaning
LOOP START	Initial loop-start position, from 1% to 98% of the held portion.
LOOP END	Initial loop-end position, from 2% to 99%, above Loop Start.
AKAI AUTO FIND	Optionally refines the initial loop locally using Akai's existing routine.
CROSSFADE	Optionally processes the join using the native crossfade function.
XFADE AMOUNT	Requested crossfade length as a percentage of the resulting loop length.
XFADE SHAPE	Selects LINEAR, LOG or SINE.
CROP AFTER LOOP	Chooses how much audio remains after a successful loop is prepared.

For a steady synth tone, try a five-second Note Time, Loop Start 50%, Loop End 95%, Auto Find ON and Crossfade ON at 10%. These are starting settings, not a promise that every patch will loop cleanly. Aim for a stable section after the initial attack and before the release.

Auto Find OFF keeps the percentage positions as the starting loop without native refinement; the recording still has to pass the loop checks. **Auto Find ON** performs a local adjustment. It does not search every possible loop throughout the entire 50–95% region or automatically identify the musically best sustain.

Crossfade amount is limited by the available pre-loop audio and loop length. A larger percentage can smooth a join but may change the sound. Audition the curves on separate test captures. With Crossfade OFF, amount and shape are hidden.

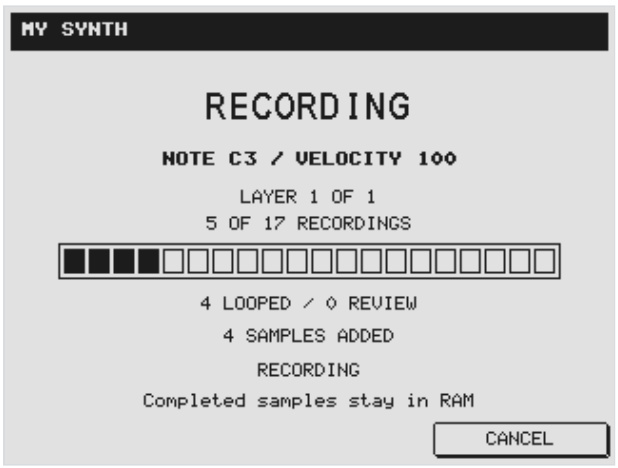
What happens to the tail?

Result	Audio retained
AUTO LOOP OFF	The ordinary recorded note and tail.
Successful loop, CROP OFF	The held-note recording; the external release tail is removed.
Successful loop, CROP ON	The attack through loop end, plus the small native 15-frame playback margin.
Loop rejected for review	The original recording, including its recorded tail.

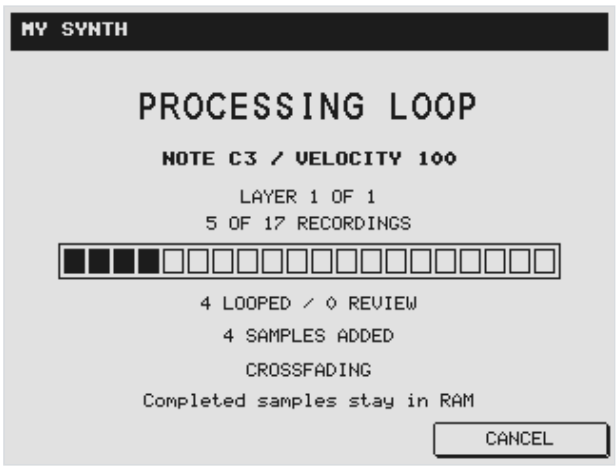
The sampler records the external release before processing the loop. Cropping does not skip that waiting period. For the finished looped program, adjust playback release using the Akai's envelope controls.

Auto Loop checks each channel and can leave a recording unlooped for review. Listen to these results in the ordinary sample editor. Evolving pads, heavy chorus, noisy textures and rhythmic material may require different positions or manual editing. Loop Setup contains settings for the recordings you are about to make. Use the ordinary sample editor to inspect their waveforms after capture.

7. Run, listen and save



Recording



Loop processing

Press **START** on the main page. The display follows preparation, waiting for audio, recording, release and sample assignment. When looping is enabled, it also shows the processing stage and counts of looped samples and recordings requiring review.

CANCEL stops the run while keeping completed sample assignments in RAM. Finishing an active recording or processing operation may take a moment. The current unfinished capture may not be assigned to the program; inspect the samples in RAM if the display asks you to check them.

After completion, play across the keygroups and velocity boundaries. For looped sounds, hold notes long enough to hear several repetitions, then check the release. Review any unlooped recordings and make ordinary sample or program edits as needed.

Save the program with its samples through the normal SAVE page. Auto Sample does not write your sound to disk. Unsaved recordings remain dependent on RAM and are lost when the sampler is powered off. Reload a saved program to confirm that the samples and loop settings were stored as intended.

8. Troubleshooting

What you notice	What to check
USE AN EMPTY PROGRAM	Create a new named program with no sample assignments. A completed or partial Auto Sample program is no longer empty.
MIDI plays, but there is no recorded audio	Check audio cables, input source, Record Mode, gain and Start Level. TEST NOTE and the meters help separate MIDI routing from audio routing.
NO AUDIO – CHECK START LEVEL	The audio did not reach the trigger threshold in time. Check the signal and threshold with a one-note test.
A recording is very quiet	Adjust audio output/recording gain while watching the meters. Check your chosen capture velocity too.
Decay is cut short	Lower End Level, increase Quiet Time or minimum timing, or use OFF to compare a full-duration recording.
The tail always uses its maximum time	Background noise may remain above End Level; in stereo, check both channels.
A loop is left for review	Try a steadier section, a longer held recording, different percentages or crossfade settings. The original recording is available for listening and manual work.
An option has disappeared	Enable its parent option: AUTO END for quiet controls, AUTO LOOP for loop controls, and CROSSFADE for amount/shape.
A run stops with a processing or restart message	Stop further runs, inspect the samples already in RAM and record the settings/message for feedback. Save completed work through the normal controls if they remain responsive.

For useful feedback, include your sampler model, mono/stereo mode, source instrument, capture and loop settings, the displayed message and what you expected to hear. A photo of the relevant screen is helpful.

9. Expanded Edition startup



S5000 EX



S6000 EX

The boot screen shows a CPU/circuit graphic, the correct **S5000 EX** or **S6000 EX** model logo, and **EXPANDED EDITION v1**. The EX lettering is outlined to match the model design.

The separate **Operating System VERSION 2.14** field is retained for compatibility. Expanded Edition v1 identifies the added feature set; it does not change the underlying Akai version identity. Use the checksum accompanying a firmware download to identify its exact file.

Home · Quick start · Installation · Compatibility · Risk disclaimer

A · Installation

The firmware image is **S56K-EX.bin**, shared by the AKAI S5000 and S6000. The boot screen selects the correct model graphic automatically and displays **EXPANDED EDITION v1**. The underlying AKAI operating-system identity remains **2.14**.

Read the flashing disclaimer before proceeding. Permanent installation carries a risk of an unusable sampler and possible repair costs. The instructions and compatibility notes supplied with your exact download take precedence over this general guide.

Before you begin

- Obtain the firmware attachment and its SHA-256 checksum from the project's release area. GitHub's automatic "Source code" archives contain documentation, not the OS.
- Confirm that the release supports your sampler and follow its installation notes. If the firmware, checksum or instructions are missing, stop rather than substituting another file.
- Back up your programs and samples to separate storage. Keep a known working OS file and readable disk; these do not guarantee recovery after a failed flash.
- Verify the BIN's SHA-256 checksum against the checksum published with that exact release.
- Use a known readable floppy disk, a reliable drive and a stable power supply. Copy **S56K-EX.bin** to the disk with no other OS BIN files present. Where possible, read the file back on the computer and verify its checksum again.

1. Load temporarily from floppy

1. Insert the prepared floppy and switch on the sampler.
2. Allow it to read the OS. At **Write Flash OS**, select **NO** to continue without permanently installing it.
3. Check the model-specific EX graphic, the **EXPANDED EDITION v1** caption and normal startup.
4. Open an empty test program, then **Program Tools → Auto Sample**. Make a short capture, audition it and check the controls you intend to use. Save wanted work using the normal **SAVE** page.

If the expected artwork or feature is absent, or anything behaves unexpectedly, stop and check the disk and exact file before considering installation. A temporary boot leaves the existing installed OS in flash; it does not prove that a flash write will succeed.

2. Install permanently from floppy

Proceed only when the temporary load and checks have succeeded and you have accepted the installation risk.

1. Boot again using the same verified floppy and firmware file.
2. At **Write Flash OS**, choose **YES** when ready to install.
3. Allow writing and verification to finish. **Do not switch off, reset or remove media during the update.** Follow the sampler's completion instructions.
4. Remove the floppy and restart when the update has completed. Confirm the EX boot artwork appears without the disk, then check ordinary operation and a short Auto Sample capture.

An update that reports an error is not a successful installation. Record the exact message and consult the instructions for that release; do not assume a repeated update or a retained backup will recover every failure.

USB and ak.Sys

USB is not required for Auto Sample or floppy installation. Windows/ak.Sys installation has not been verified on physical hardware for the documented version, so this guide does not provide a USB installation procedure. Temporary USB boot without writing flash is not an advertised feature.

See Compatibility for the distinction between software checks, earlier device results and the exact version's hardware evidence.

B · Compatibility

S56K Expanded Edition v1 is designed for the **AKAI S5000 and S6000**, using their shared v2.14 operating-system foundation. One model-aware firmware image supplies the features and the appropriate startup artwork for each sampler.

Auto Sample uses the sampler's MIDI outputs and recording inputs. USB is not required for the sampling workflow. The visible **EXPANDED EDITION v1** label identifies this project's additions; the underlying OS/settings/protocol identity remains **2.14**.

Verification evidence

Area	Evidence and limits
Auto Sample on S5000 and S6000	Successful operation was reported on both models during feature development.
Permanent installation from floppy	A successful flash and subsequent reboot were reported for an earlier corrected installer build.
Current EX artwork image	Passed all 15 software verification suites and a scoped independent review. The latest corrected logos and caption still await physical confirmation on both models.
Floppy and USB flash routines	Passed bounded software checks covering writing, read-back verification and restart behavior. This does not establish every real device or host configuration.
Windows / ak.Sys USB installation	End-to-end installation on hardware remains unverified. USB boot without installation is not a supported workflow.

A successful earlier build does not certify every later image. Firmware downloads are identified by a SHA-256 checksum; consult the installation notes and verification status accompanying the exact file you use. The software checks exercise native routines with controlled inputs and device models, rather than a complete hardware emulator. No installation is guaranteed risk-free; read the risk disclaimer.

Before a capture session

- Create a new, empty program and name it before opening Auto Sample.
- Leave generous free sample RAM. Longer recordings, stereo, additional velocities and temporary loop-processing copies need more space.
- Test the input level and attack on one note before capturing a large range.
- Avoid simultaneous sample editing through ak.Sys or SysEx while Auto Loop is running.
- Save the program **with its samples** through the normal SAVE page before switching off.

Auto Loop is optional. Its native processing can prepare useful sustain loops, but results depend on the sound. Recordings that fail the normal loop checks retain their original audio for review. See the loop guide for cropping and release-tail behavior.

Loading and installation

The installation guide separates temporary floppy loading from permanent flash installation. The established first check is to boot from a readable floppy and select **NO** at the “Write Flash OS” prompt. Confirm operation before considering a permanent update.

Firmware downloads are withheld from this repository at present. See Release information for availability. The guide and screenshots are documentation, not an alternative firmware download.

About the images

The screen gallery uses native firmware renders at the original **320 × 240** resolution. Program names, settings and progress figures are examples; actual physical LCD appearance varies.

The landing-page studio image is presentation artwork. It illustrates the two sampler models and the workflow, rather than documenting a hardware test or measured audio result.

C · Flashing disclaimer

Installing or using S56K Expanded Edition is your choice and is undertaken at your own risk. Read this notice and the instructions supplied with the specific firmware release before loading or installing it.

Understand the risk

S56K Expanded Edition is an independent modification for vintage AKAI S5000 and S6000 samplers. Writing firmware to the sampler's flash memory changes its installed operating system. An interrupted or unsuccessful update, incompatible hardware, failing media, power loss, or an undiscovered software fault may leave the sampler unable to start or operate correctly. Recovery may require specialist equipment, repair, replacement parts or service, and may not be possible.

Loading firmware temporarily and obtaining a successful result do not guarantee that a later flash operation will succeed. Testing, checksums and retained backup files reduce some risks; they do not make an update risk-free or provide automatic recovery.

Your responsibility

You are responsible for deciding whether to use the firmware, checking the applicable model and release instructions, verifying the downloaded file, backing up programs and samples, using suitable media and a reliable power supply, and determining whether you have the knowledge and resources needed to proceed. Do not interrupt an update once flash writing has started. If you are uncertain about the process or cannot accept the possibility of loss or repair costs, do not install it.

No warranty

To the maximum extent permitted by applicable law, the firmware, documentation and related materials are provided **“as is” and “as available,” without warranties of any kind**, whether express, implied or statutory. This includes, where applicable, warranties of merchantability, fitness for a particular purpose, satisfactory quality, accuracy, non-infringement and freedom from defects. No promise is made that the software will be error-free, meet your requirements, work with every hardware configuration, or allow recovery after a failed update.

Limitation of liability

To the maximum extent permitted by applicable law, the project owner, authors, maintainers, contributors and distributors will not be liable for any loss or damage arising from downloading, loading, installing, using or being unable to use the firmware or related materials. This includes damage to or loss of use of equipment; loss or corruption of programs, samples or other data; repair, replacement or recovery costs; loss of income or business; and direct, indirect, incidental, special or consequential loss, whether the claim arises in contract, tort or otherwise, even if the possibility of such loss was known.

Nothing in this notice excludes or limits liability that cannot lawfully be excluded or limited, or affects rights that cannot be waived under applicable law. Where a limitation is not permitted, it applies only to the extent the law allows.

Independent project

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